Types of functions. Part 1

In this session we will revisit the types of functions and their main characteristics, such as domain, range, asymptotes, and holes[1]. Furthermore, we will also review the translation of functions. This is becuase by knowing the domain and range of these types of fuctions is going to help in identifying the domain and range of the derivatives.

[1] January 16

Polynomials For polynomials of the form[2]:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0$$

where n is a nonnegative integer.

Polynomial Type	**Domain**	**Range**
Odd degree (n odd)	$(-\infty, \infty)$	$(-\infty, \infty)$
Even degree (n even)	$(-\infty, \infty)$	Depends on leading coefficient:
$a_n > 0$		$[k, \infty)$
$a_n < 0$		$(-\infty, k]$

[2] **Notes:**

- Domain of *all* polynomials is all real numbers, $\mathbb{R} = (-\infty, \infty)$.
- For odd-degree polynomials, the range is always all real numbers due to the end behavior: as $x \to \pm\infty$, $f(x) \to \pm\infty$ (or $\mp\infty$ if $a_n < 0$).
- For even-degree polynomials:
  - If $a_n > 0$, the graph opens upward with a **minimum value** k.
  - If $a_n < 0$, the graph opens downward with a **maximum value** k.
  - k is found using calculus techniques (derivative = 0) or vertex formulas for quadratics.

Exponential Functions For functions of the form[3]:

$$f(x) = a^x \quad \text{or} \quad f(x) = b \cdot a^{kx} + c$$

where $a > 0$, $a \neq 1$, and $b \neq 0$.

Function Type	**Domain**	**Range**
Basic: $f(x) = a^x$	$(-\infty, \infty)$	$(0, \infty)$
With vertical shift: $f(x) = a^x + c$	$(-\infty, \infty)$	(c, ∞)
General: $f(x) = b \cdot a^{kx} + c$	$(-\infty, \infty)$	If $b > 0$: (c, ∞) If $b < 0$: $(-\infty, c)$

[3] **Special Cases**

- **Natural exponential**: $f(x) = e^x$ has domain $\mathbb{R}$ and range $(0, \infty)$.
- **Natural logarithm**: $f(x) = \ln(x)$ has domain $(0, \infty)$ and range $\mathbb{R}$.
- **Exponential with base** $0 < a < 1$: Domain is still $\mathbb{R}$, range is $(0, \infty)$ if no vertical shift.
- **Key relationship**: $f(x) = a^x$ and $g(x) = \log_a(x)$ are inverse functions:

$$\text{Domain of } f = \text{Range of } g = \mathbb{R}$$

$$\text{Range of } f = \text{Domain of } g = (0, \infty)$$

Calculus Connection

- For $f(x) = a^x$, the derivative is $f'(x) = a^x \ln(a)$
- For $f(x) = \ln|x|$, the derivative is $f'(x) = \frac{1}{x}$ (for $x \neq 0$)
- Domain restrictions are crucial when finding derivatives: $\frac{d}{dx}[\ln(u(x))] = \frac{u'(x)}{u(x)}$ only when $u(x) > 0$

Logarithmic Functions For functions of the form:

$$f(x) = \log_a(x) \quad \text{or} \quad f(x) = b \cdot \log_a(kx + d) + c$$

where $a > 0$, $a \neq 1$, and $b \neq 0$.

Function Type	Domain	Range
Basic: $f(x) = \log_a(x)$	$(0, \infty)$	$(-\infty, \infty)$
With horizontal shift: $f(x) = \log_a(x-h)$	(h, ∞)	$(-\infty, \infty)$
General: $f(x) = b \cdot \log_a(kx+d) + c$	Solve: $kx + d > 0$ Typically: $(-\frac{d}{k}, \infty)$	$(-\infty, \infty)$

Types of functions. Part 2

Rational functions For rational functions[4] of the form[5]:

$$f(x) = \frac{P(x)}{Q(x)}$$

where $P(x)$ and $Q(x)$ are polynomials, $Q(x) \neq 0$. A basic example: $f(x) = \frac{1}{x}$

Function	Domain	Range
$f(x) = \dfrac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$

- **Vertical asymptote** at $x = 0$
- **Horizontal asymptote** at $y = 0$
- **Behavior**: As $x \to 0^-$, $f(x) \to -\infty$; as $x \to 0^+$, $f(x) \to \infty$

  Here are some common patterns and examples

Example	Domain	Range
$f(x) = \dfrac{1}{x^2}$	$(-\infty, 0) \cup (0, \infty)$	$(0, \infty)$
$f(x) = \dfrac{x}{x^2+1}$	$(-\infty, \infty)$	$[-\frac{1}{2}, \frac{1}{2}]$
$f(x) = \dfrac{x^2-1}{x^2-4}$	$x \neq \pm 2$	$(-\infty, \frac{1}{4}] \cup (1, \infty)$
$f(x) = \dfrac{x-1}{x^2-1}$	$x \neq \pm 1$ (hole at $x = 1$)	$(-\infty, 0) \cup (0, \infty)$

[4] January 19

[5] **Finding Domain and Range: Step-by-Step**

1. **Domain**:
   (a) Factor numerator and denominator completely
   (b) Set denominator $Q(x) = 0$
   (c) Domain: All real numbers except solutions to $Q(x) = 0$
   (d) If factors cancel (holes), note them separately
2. **Range** (requires calculus for certainty):
   (a) Find horizontal asymptotes: Compare degrees of $P(x)$ and $Q(x)$
   (b) Find vertical asymptotes: Zeros of denominator (not cancelled)
   (c) Find critical points: Solve $f'(x) = 0$ or undefined
   (d) Analyze behavior at asymptotes and critical points
   (e) Consider end behavior as $x \to \pm\infty$